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India SMART UTILITY Week 2026
10 – 14 March 2026, New Delhi

Role of Small Modular Reactors (SMR) in The Net Zero Power Systems

Electron Accelerator driven Thorium based Molten Salt Micro Modular Reactor (eMMR)

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INTRODUCTION

Micro Modular Reactor (MMR) – a subcategory of Small Modular Reactors (SMR) capable of generating up to 10 MW.

Features:

- Electron ADSR (Accelerator Driven Subcritical Reactor)
- Molten salt fuel and coolant
- Neutron production through high energy photons
- Criticality close to 1
- Thorium fuel to minimise actinides and to extend fuel life through breeding

Problems and solutions:

Radioactive waste with SMR
Solution - Thorium fuel, reducing minor actinide production by approximately 100-fold compared to uranium-based fuels.

Large and expensive proton machines
Solution - Electron accelerator, smaller and cheaper

Licencing and deployment
Solution - Consideration of a notably lower power output (1 MW)

Simulation programs and design model

- EGS
- Geant4
- MCNPX

Studies of EM showers in molten salt with various additives – EGS program and Geant4

Neutron production - Geant4

The criticality of the reactor – MCNPX

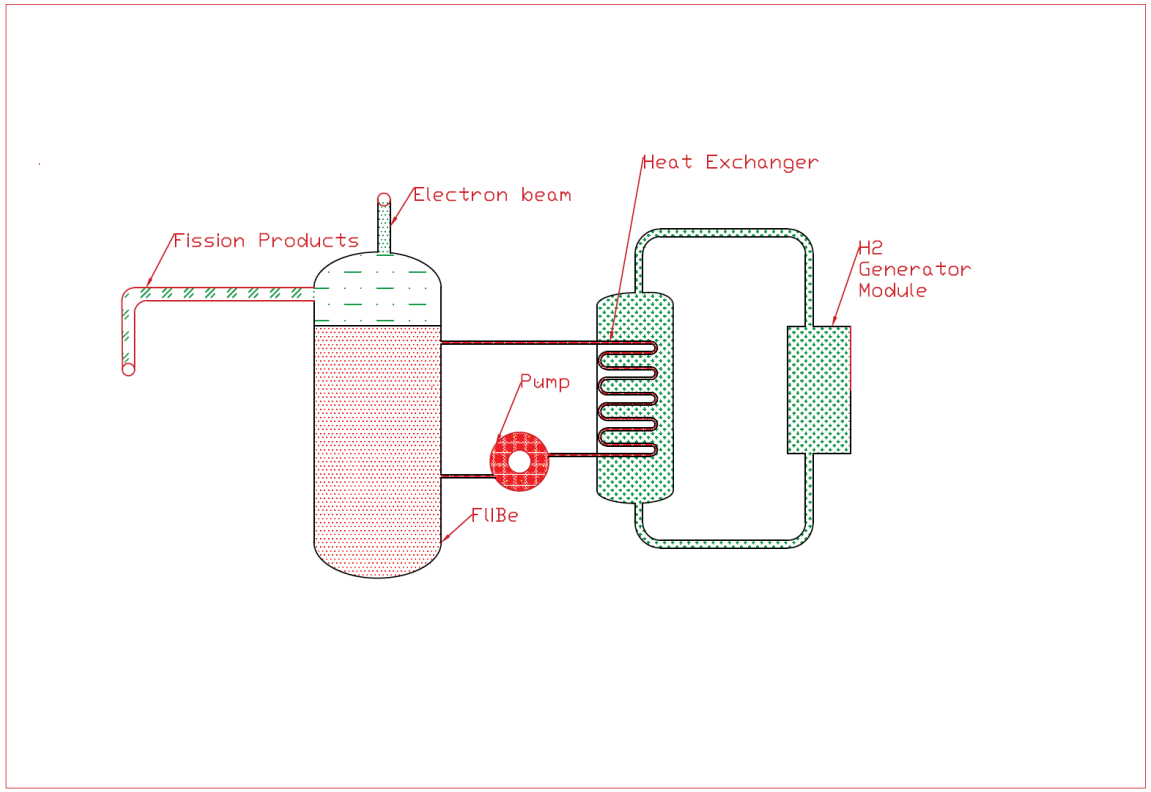

Figure 1

Neutron production - two – stage process:

1) The electrons produce an electromagnetic (EM) shower through Bremsstrahlung (e− → e− γ) and pair production (γ → e+ e−)

2) The gamma component produces neutrons off target nuclei, chiefly through the giant dipole resonance.

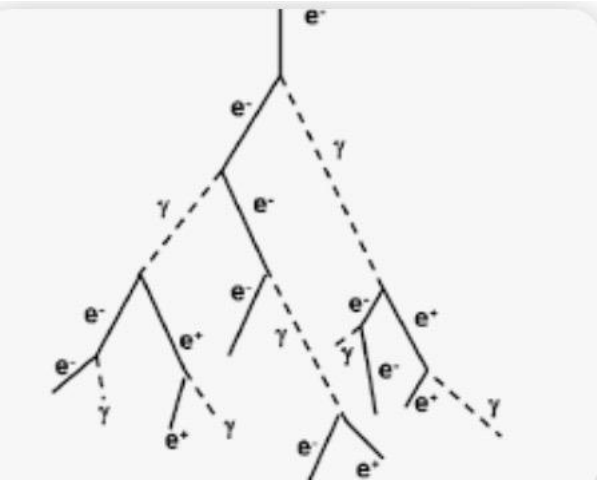

Figure 2

Molten salt system (Li_2BeF_4) containing dissolved fissile ^{233}U and fertile ^{232}Th.

- Neutrons trigger fission in ^{233}U, leading to a subcritical chain reaction that amplifies energy production.

- As ^{233}U is consumed, it will be replenished through breeding from ^{232}Th.

- The resulting heat can be harnessed for electricity generation or process heat.

Photoneutrons and their energy distribution

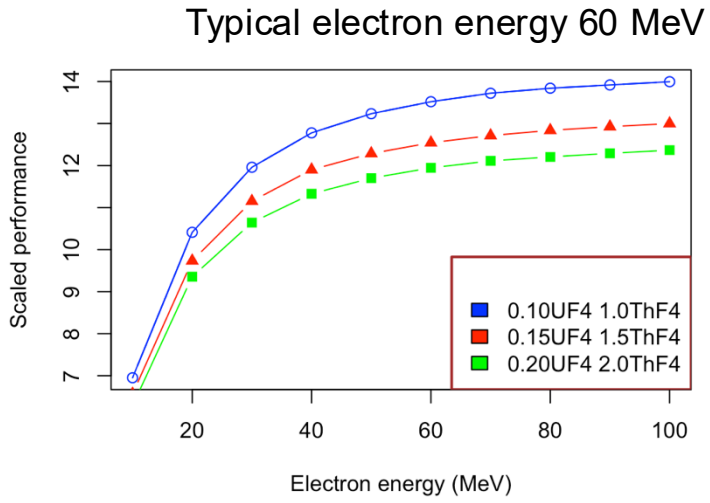

Figure 3

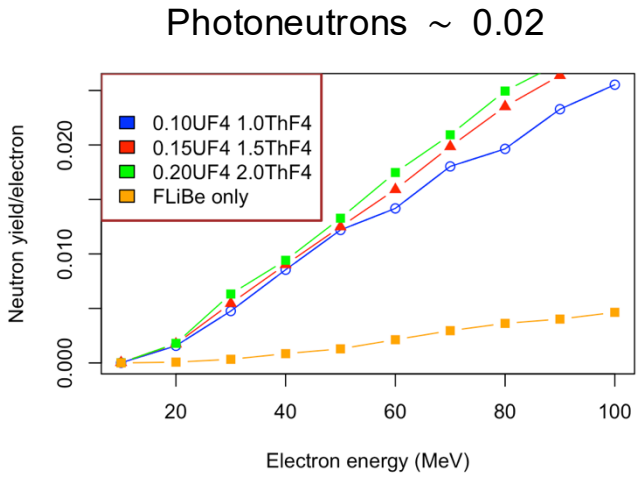

Figure 4

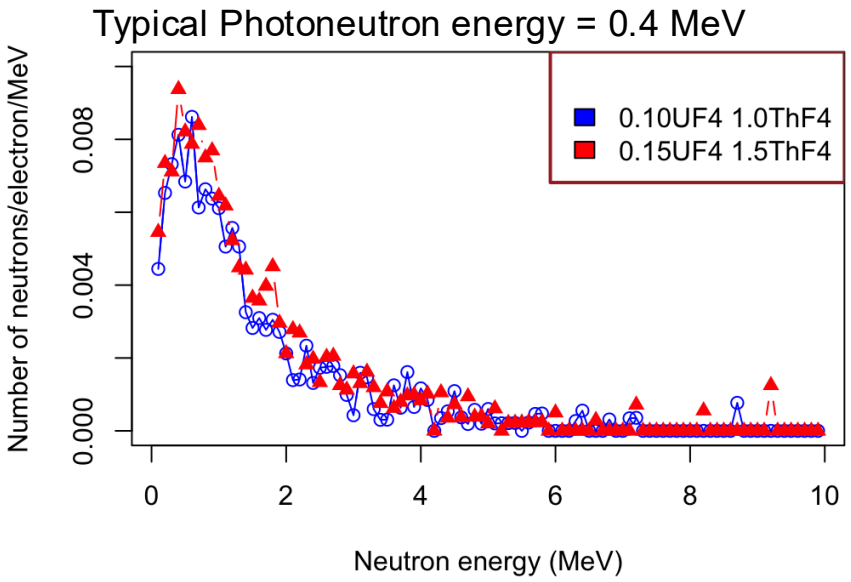

Figure 5

Criticality, breeding and two reactor system

Thorium decreases the criticality

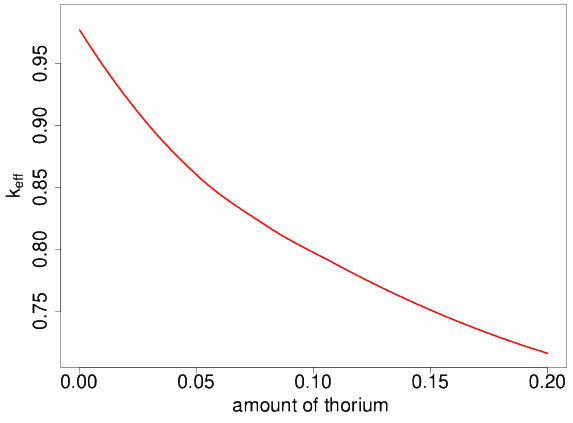

Figure 6

Isotopically purified lithium for improved neutronics

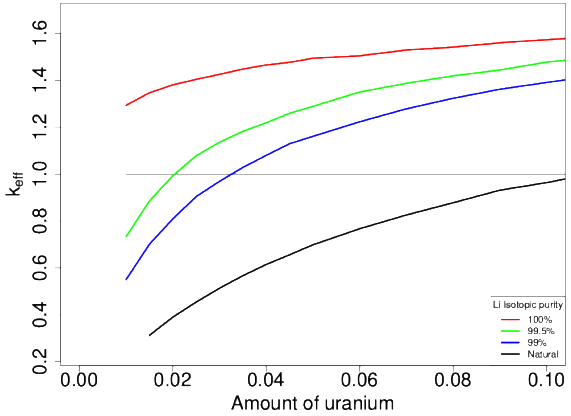

Figure 7

Coupled reactor system

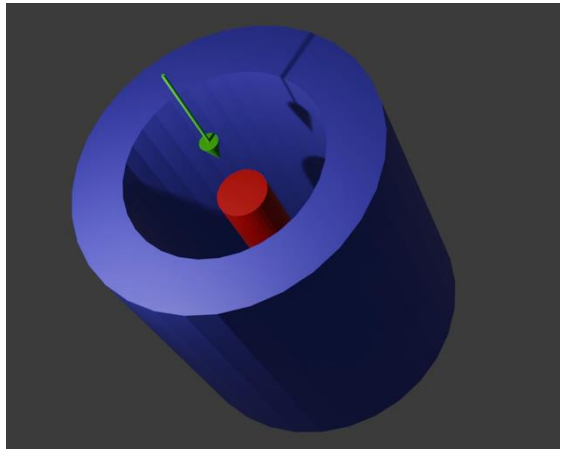

Figure 9

- Two reactors with asymmetric coupling
- Multiplication M=M1 x M2 but overall criticality limited

Breeding - As ^{233}U is consumed, it will be replaced using the fertile-to-fissile conversion.

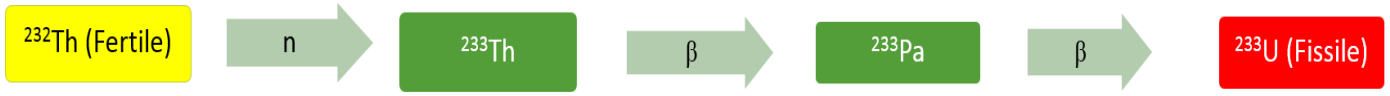

Figure 8

KEY TAKEAWAYS / RECOMMENDATIONS

- Micro modular **sub-critical molten salt breeder reactor** design.
- Driven by an **electron accelerator**.
- **Fuel:** ^{233}U seed; replaced by **thorium breeding** during operation.
- **Criticality:** ~0.9–1.0 possible with thorium in FLiBe salt.
- Fuel mix: **~1 ^{232}Th atom + 0.1 ^{233}U per FLiBe unit**.
- **Lithium purification needed** (remove ^{6}Li due to high neutron absorption).
- **Optimal electron energy:** ~60 MeV.
- Produces **≈0.02 photoneutrons per beam particle**.
- Near-criticality target: **k ≈ 0.98**.
- **Power output** depends on beam power + reactor criticality.
- **Safety challenge**; may require **two-stage coupled system**.

References

1. *H Nifenecker, S David, JM Loiseaux, and O Meplan. Basics of accelerator driven subcritical reactors. Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, 463(3):428-467, 2001.*

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